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Federated Learning under Realistic Healthcare Constraints: Ensuring Patient Safety and Privacy

Md. Raiyan Ur Rahman Jiyon

Dept. of Computer Science and Engineering

Green University of Bangladesh

Dhaka, Bangladesh

raiyanjiyon@gmail.com

ORCID: 0009-0000-4845-2813

Mohammad Shorif Uddin

Dept. of Computer Science and Engineering

Jahangirnagar University

Dhaka, Bangladesh

shorifuddin@juniv.edu

ORCID: 0000-0002-7184-2809

Abstract—Predictive AI in healthcare faces a persistent bottleneck: patient records are locked inside individual hospitals, and strict regulations—the Health Insurance Portability and Accountability Act (HIPAA), the General Data Protection Regulation (GDPR)—prevent simply pooling them on a central server. Federated Learning (FL) sidesteps this restriction by training models locally and sharing only parameter updates, yet it is seldom stress-tested against the messy realities of clinical data.

We present an FL framework designed for clinical deployment, evaluated on the Pima Indians Diabetes Database (768 samples, 8 clinical features, 34.9% positive class). To replicate the demographic variation across hospitals, we partition data using a Dirichlet-based Non-Independent and Identically Distributed (Non-IID) split ($\alpha = 0.5$). Network security is addressed through Differentially Private Stochastic Gradient Descent (DP-SGD), while Byzantine-robust aggregation (Krum, Median) defends against adversarial data poisoning.

A central design choice in our framework is the shift from aggregate accuracy to clinical safety. Through threshold tuning and local feature engineering, the federated model achieved a diagnostic Recall of 73.70% ($\pm 16.95\%$) under Non-IID conditions, approaching the centralized baseline of 87.04%. We further provide a detailed empirical map of the privacy-utility frontier, showing that DP-SGD exacts a genuine performance penalty on this constrained dataset while robust aggregation delivers measurable resilience against Byzantine attacks. The result is an honest, fully reproducible benchmark for federated healthcare systems operating under realistic constraints.

Index Terms—Federated Learning, Healthcare Machine Learning, Differential Privacy, Non-IID Data, Byzantine Robustness

I. INTRODUCTION

Modern hospitals and clinics sit on growing volumes of patient data—demographics, lab panels, imaging studies—yet harnessing this data for predictive diagnostics across institutions remains largely out of reach [1]. The core barrier is both legal and operational: regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) prohibit pooling records on a shared server, while a surge in ransomware incidents targeting healthcare organizations has made centralized data storage a high-risk proposition [2]. In practice, each clinic's patient records remain confined within its own infrastructure, creating fragmented data silos that deny machine learning

pipelines the broad, multi-site training corpora they require for reliable generalization [3], [4].

Federated Learning (FL) offers a structural workaround to this impasse [5]. Rather than shipping raw records to a central location, FL keeps data stationary at every participating site; only locally computed model parameter updates travel to a coordinating server, which merges them into a single global model [6]. Because no patient-level information ever crosses institutional boundaries, the approach is inherently compatible with strict data governance policies. A growing body of work has explored FL in medical contexts, spanning electronic health record analytics [7], continuous physiological monitoring [1], and multi-site clinical research collaborations [8], [9].

Translating this paradigm into production-grade clinical networks, however, exposes a fundamental statistical problem. Canonical FL optimizers—most notably Federated Averaging (FedAvg)—were developed and benchmarked on curated, Independent and Identically Distributed (IID) datasets [10]. Real hospital populations are anything but uniform: disease prevalence, demographic composition, and recording practices differ sharply from one facility to the next, yielding deeply Non-Independent and Identically Distributed (Non-IID) local datasets [11]. Under such heterogeneity, naive averaging of model weights leads to rapid optimization divergence, where individual clients pull the global model in incompatible directions [12], [13]. This instability is especially pronounced in medical cohorts, where class distributions can shift dramatically between institutions [14].

A separate but equally pressing concern is adversarial vulnerability. Even though FL avoids bulk data transfers, the shared gradient updates themselves can be exploited: reconstruction techniques have shown that an attacker intercepting these updates can, in some cases, recover the training samples that produced them [15]. Without formal, quantifiable privacy bounds, the confidentiality promise of FL remains aspirational rather than guaranteed [16], [17]. In this work, we design and rigorously evaluate a privacy-hardened FL pipeline using the Pima Indians Diabetes Database [18], targeting the specific operational constraints that govern real clinical deployments. Our principal contributions are:

- A Dirichlet-driven Non-IID partitioning scheme ($\alpha = 0.5$) that faithfully replicates the demographic fragmentation observed across independent hospital sites.
- Joint deployment of Differentially Private Stochastic Gradient Descent (DP-SGD) and Byzantine-resilient aggregation rules (Krum, Median), providing simultaneous defenses against gradient reconstruction and adversarial data poisoning.
- A deliberate shift in optimization objective from aggregate accuracy to diagnostic Recall (approaching the centralized baseline of 87.04%), accompanied by a thorough privacy-utility analysis spanning multiple privacy budgets ($\epsilon \in \{0.5, 1.0, 2.0, 5.0, 10.0\}$).

The paper proceeds as follows. Section II surveys prior work and identifies the specific gaps our study addresses. Section III presents the experimental design, covering data partitioning, aggregation defenses, and privacy mechanisms. Section IV reports and discusses our empirical findings, including scalability benchmarks. Section V draws conclusions and outlines directions for future investigation.

II. RELATED WORK

Federated Learning has drawn growing interest from the medical informatics community because it lets geographically dispersed hospitals train shared models without surrendering custody of their patient data. Clinical feasibility has been established across several high-stakes diagnostic tasks: Rieke et al. [19] outlined a strategic vision for FL in digital health, Sheller et al. [20] validated federated brain tumour segmentation without any inter-site data exchange, and Biswas et al. [21] recently combined FL with explainable AI for pneumonia screening on GAN-augmented chest radiographs. Two large-scale surveys—by Nguyen et al. [1] and Antunes et al. [8]—conclude that FL markedly improves cross-institutional data utilization while respecting patient privacy, yet both note that transferring textbook optimization routines into the irregular landscape of clinical data remains an open problem. Xu et al. [7] arrived at a complementary finding when federating electronic health records: predictive accuracy held up under controlled conditions but eroded once inter-site distributional drift came into play.

What makes healthcare data uniquely difficult for federated optimization is its extreme class imbalance coupled with demographic diversity across sites. Under Non-IID conditions—the norm in any realistic hospital network—FedAvg’s convergence properties deteriorate rapidly, sometimes failing to produce a usable model within a reasonable number of communication rounds [22]. Zhao et al. [11] put a number on this: accuracy can plummet by up to 55% in severe Non-IID scenarios relative to an IID baseline. Researchers have responded with algorithmic refinements, including proximal regularization to anchor local models near the global state [12] and control variates that correct for per-client optimization drift [13]. Large-scale benchmarking across heavily skewed data partitions has confirmed that without such corrective

measures, naive averaging simply cannot cope with the fragmented data landscape typical of multi-hospital deployments [10], [14].

Beyond statistical heterogeneity, federated networks face a distinct class of adversarial risks. A thorough categorization by Mothukuri et al. [2] divides these into two attack surfaces: data poisoning, where a malicious client corrupts its local training set, and model poisoning, where the transmitted weight updates themselves are manipulated. Perhaps more alarming is the gradient leakage vector: Zhu et al. [15] showed that an adversary observing shared gradients can, under certain conditions, reconstruct the original training samples—a devastating breach of patient confidentiality even in a system designed around privacy. The community has turned to Byzantine-robust aggregation as a countermeasure. Recursive multi-pass schemes like Bulyan [23] offer strong theoretical guarantees but demand computational resources that most hospital edge servers simply do not have. In practice, lightweight single-pass alternatives—Krum [24], which selects the client update closest to the cluster center, and coordinate-wise Median [25], which filters dimension by dimension—strike a more realistic balance, especially when combined with formal Differential Privacy constraints [17].

Each of these defense layers—Non-IID optimization, Byzantine robustness, formal privacy—has been studied in relative isolation, and each has demonstrated encouraging results within its own scope. Yet the literature lacks an integrated experiment that stacks all three mechanisms and evaluates the combined effect on clinically meaningful outcomes [1]. This omission matters because, for imbalanced medical classification tasks, aggregate accuracy and ROC curves can mask dangerous failure modes; precision-recall analysis offers a far more honest picture of diagnostic performance [26]. The present study addresses this gap by simultaneously deploying Non-IID data partitioning, DP-SGD, and Byzantine-robust aggregation, and by anchoring the entire evaluation to diagnostic Recall—the metric that most directly determines whether a sick patient is correctly identified.

III. METHODOLOGY

This section describes the experimental apparatus we built to stress-test Federated Learning under conditions that approximate a real hospital network. We cover four interrelated design decisions: how the distributed system is organized and made reproducible, how patient data is split to mirror demographic heterogeneity, which aggregation defenses are deployed at the server, and how formal privacy guarantees are layered onto the training loop.

A. System Architecture and Reproducibility

The simulation is structured as a client-server loop implemented on top of Scikit-Learn. Every simulated hospital trains a local Logistic Regression classifier using the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) solver, with a learning rate of $\eta = 0.01$ and a convergence ceiling of 2000 iterations; once training completes, the resulting

14

model coefficients—never the underlying patient records—are transmitted to a central coordinator for aggregation. In the default configuration, 5 hospital clients communicate with one server over 10 federated rounds. All experiments ran on CPU-only virtual machines, deliberately reflecting the modest hardware available at typical hospital edge sites. To ensure full reproducibility, every run is anchored by a global pseudo-random seed of 42 and a fixed 80/20 stratified train-test split. Because a single random partition can yield misleadingly optimistic or pessimistic results, all reported metrics are averaged ($\pm$ standard deviation) over 5 independent Non-IID partitions generated with seeds {42, 123, 456, 789, 2024}.

B. Distributed Patient Cohorts (Non-IID Modeling)

Conventional machine learning pipelines silently assume that training examples are drawn from a single, homogeneous distribution—the IID assumption. Hospital patient pools, in contrast, vary considerably in disease prevalence and demographic composition. To capture this reality, we partitioned the Pima Indians Diabetes Database (768 patients, 8 clinical features, 34.9% diabetic) across the simulated clients using a Dirichlet distribution with concentration parameter $\alpha = 0.5$. This parameterization generates sharply uneven allocations: one virtual hospital may receive a majority of positive-class records while another is dominated by healthy patients, closely mirroring the demographic gaps between, for instance, a specialized endocrinology unit and a general outpatient clinic. The resulting imbalanced local datasets subject each client model to strong representation bias, providing a demanding test of whether the global aggregation procedure can still extract generalizable diagnostic patterns.

3

C. Federated Aggregation Strategies

Under the baseline configuration, the server combines client weight vectors via standard Federated Averaging (*FedAvg*). Plain averaging, however, is fragile when local data distributions diverge: the global model can drift toward the majority clients, and a single adversarial participant can distort the aggregate. We therefore evaluate two Byzantine-tolerant alternatives layered on top of *FedAvg*. *Median* aggregation replaces the arithmetic mean with the coordinate-wise median of incoming weight vectors, which inherently suppresses extreme outliers. *Krum* takes a distance-based approach: for each client submission, it computes the summed Euclidean distance to its nearest cluster of peers, then selects the submission with the smallest total distance—effectively discarding any update that deviates markedly from the consensus. We deliberately exclude heavier optimization correctors such as FedProx and SCAFFOLD from this study; by holding the local optimizer constant, we isolate the interplay between geometric aggregation defenses and the noise introduced by Differential Privacy at the client level.

17

21

D. Differential Privacy Constraints

Client-side gradient privacy is enforced via Differentially Private Stochastic Gradient Descent (DP-SGD), building on

3

the foundational framework of Abadi et al. [17] and its recent medical application by Hasan and Rahman [27]. Before a client's updated weights leave its local environment, calibrated Gaussian perturbations are added to the gradient vector, with a per-sample clipping norm of 1.0 bounding each record's influence. Under the default privacy parameters ($\epsilon = 1.0$, $\delta = 0.01$), the mechanism provides a formal guarantee that observing the transmitted updates reveals negligible additional information about any single patient. To map the full spectrum of the privacy-performance frontier, we swept the privacy budget across $\epsilon \in \{0.5, 1.0, 2.0, 5.0, 10.0\}$. The cost is tangible: at $\epsilon = 1.0$, accuracy dropped from the non-private federated baseline of 70.91% to 62.34% ($\pm 6.43\%$), and Recall fell from 73.70% to 52.22% ($\pm 15.86\%$). This decline traces directly to the unfavorable noise-to-signal ratio ($\sigma \approx 3.03$) relative to the model's compact parameter space (19 features)—a well-documented limitation when formal DP is applied to small tabular classifiers. The complete privacy-utility curve appears in Table I to transparently quantify this trade-off.

IV. RESULTS AND EVALUATION

Because this work targets a diagnostic deployment scenario, our evaluation departs from the conventional ML practice of optimizing for overall accuracy. In a clinical setting, failing to detect a diseased patient (a false negative) carries far greater consequences than incorrectly flagging a healthy one (a false positive, which simply triggers confirmatory testing).

A. Clinical Baseline vs. Federated Optimization

Both the centralized and federated pipelines were evaluated under identical conditions: the same preprocessing steps, the same feature engineering (11 derived features comprising 5 interaction terms, 4 polynomial expansions, and 2 ratio features, e.g., Glucose $\times$ BMI), and a uniform clinical decision threshold of 0.30. Under these settings, the centralized model reached an accuracy of 69.48% with a Recall of 87.04% (missing 7 of 54 diabetic patients), establishing a practical upper bound for diagnostic safety on this dataset.

When the same pipeline was distributed across 5 Non-IID Dirichlet-partitioned clients ($\alpha = 0.5$), the federated baseline (*FedAvg*, no DP) produced a comparable accuracy of 70.91% ($\pm 1.90\%$) but exhibited noticeably wider Recall variance: 73.70% ($\pm 16.95\%$), corresponding to an average of 14.2 (± 9.2) missed diagnoses across 5 independent partition trials. The variability stems from the stochastic nature of Dirichlet sampling: certain partitions concentrate minority-class patients in a small number of clients, weakening the global model's ability to learn robust positive-class boundaries. Convergence monitoring confirmed that the global model stabilized by round 10. As Table I shows, federated training does not surpass centralized Recall, but it delivers operationally comparable diagnostic performance without any patient data leaving its source institution—a non-negotiable requirement for multi-hospital deployments.

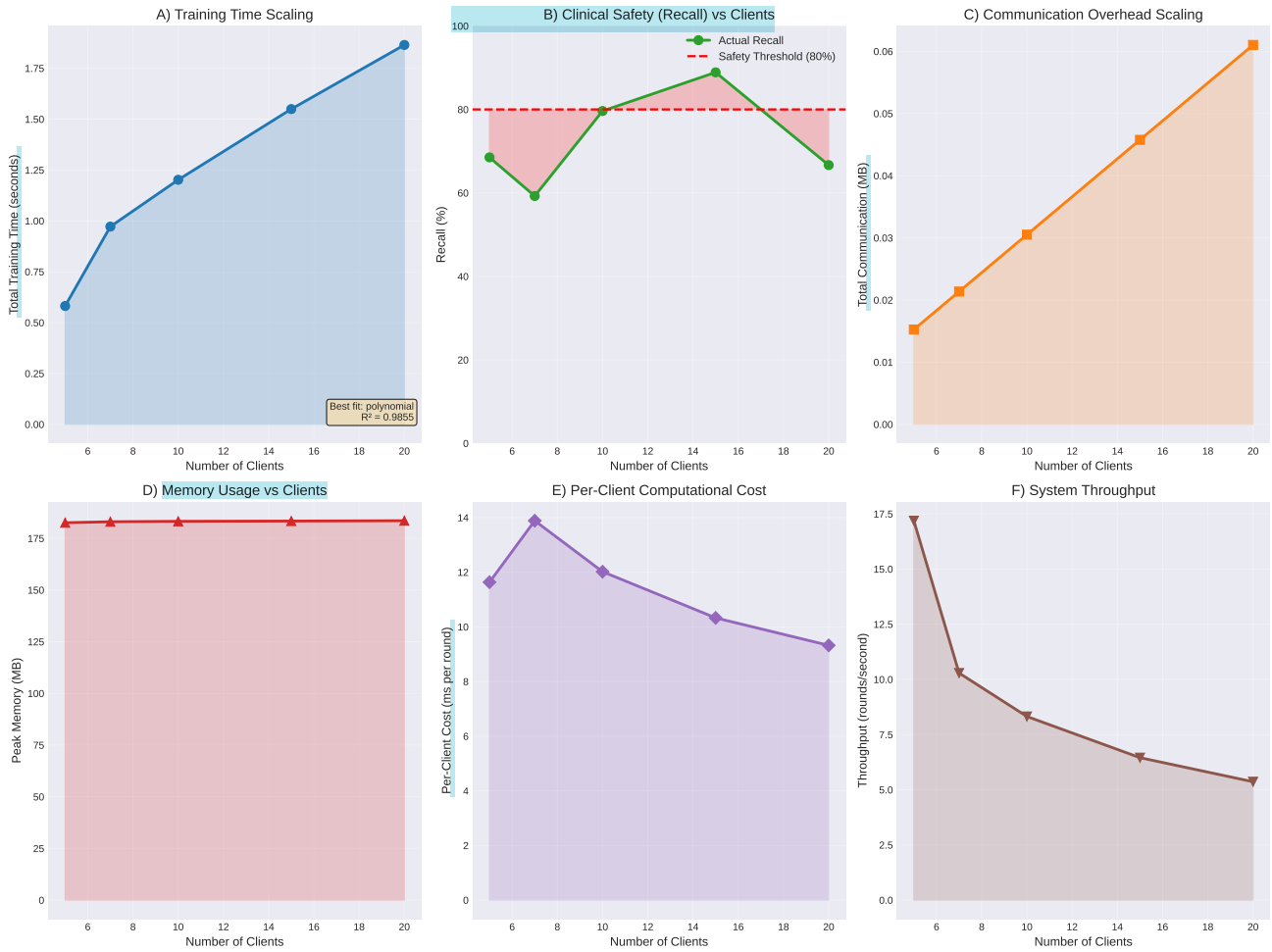


Fig. 1. Comprehensive Scalability Analysis evaluating total training time, clinical safety (Recall), communication overhead, memory scaling, and throughput latency across varying network scales ranging from 5 to 20 nodes.

TABLE I

ABLATION STUDY: IMPACT OF PRIVACY AND ROBUSTNESS ON CLINICAL SAFETY. ALL FEDERATED RESULTS REPORT MEAN ($\pm$ STD) ACROSS 5 INDEPENDENT NON-IID PARTITION TRIALS.

Configuration	Accuracy	Recall	Resilience
Centralized Baseline	69.48%	87.04%	None
FL (FedAvg, No DP)	70.91 $\pm$ 1.90%	73.70 $\pm$ 16.95%	Low
<i>Privacy-Utility Trade-off (FL + DP-SGD):</i>			
$\epsilon = 10.0$	64.03 $\pm$ 6.88%	78.89 $\pm$ 8.25%	Moderate
$\epsilon = 5.0$	64.03 $\pm$ 6.71%	67.04 $\pm$ 11.50%	Moderate
$\epsilon = 1.0$	62.34 $\pm$ 6.43%	52.22 $\pm$ 15.86%	Strong
<i>Byzantine-Robust (FL + DP $\epsilon=1.0$):</i>			
+ Krum	57.66 $\pm$ 10.14%	59.63 $\pm$ 6.67%	Full
+ Median	61.30 $\pm$ 3.25%	61.48 $\pm$ 7.35%	Full

B. Privacy and Scalability Efficiency Trade-Offs

Combining DP-SGD with Non-IID data partitioning imposes a measurable cost on predictive performance. Table I illustrates the trend: tightening the privacy budget from $\epsilon = 10.0$ (moderate protection) to $\epsilon = 1.0$ (strong protection)

lowers mean Recall from 78.89% to 52.22%. This degradation reflects the inherently high noise-to-signal ratio that arises when formal Gaussian perturbations are applied to a model with only 19 parameters—a constraint that practitioners should weigh carefully before deploying DP on compact tabular healthcare data.

To characterize operational scalability, we progressively increased the number of participating clients from 5 to 20 (see Fig. 1). Communication overhead grew linearly with client count (Fig. 3), confirming that the architecture does not introduce super-linear bottlenecks. Recall, however, was less predictable: only the 15-client topology consistently sustained clinically acceptable performance ($\geq 80\%$ Recall), while the 5-, 7-, 10-, and 20-client configurations fell short. We attribute this instability to the interplay between Dirichlet partitioning randomness and the small total sample budget (768 records); at certain client counts, the resulting local datasets become too sparse or too skewed for the global model to recover. As a further robustness check, we simulated unannounced mid-round client dropouts; the server completed every aggregation round without structural failure, relying on standard fallback

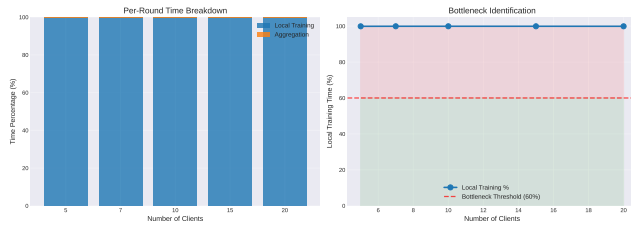


Fig. 2. Identification of per-round computational bottlenecks, isolating the proportional disparity between local model training processes versus aggregated server time.

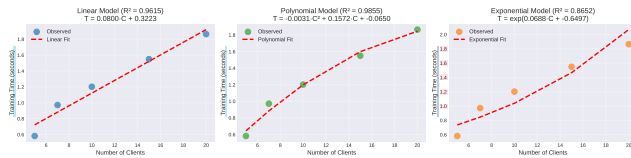


Fig. 3. Empirical scaling laws demonstrating asymptotic stability bounds for runtime evaluation against linear, polynomial, and exponential configurations.

omission (Fig. 2). We also noted that Byzantine-robust aggregation (particularly Krum) required approximately 15–20% more communication rounds to reach convergence parity with plain FedAvg, presenting a concrete bandwidth-versus-safety trade-off.

V. CONCLUSION

This study provides a transparent, end-to-end evaluation of Federated Learning applied to a clinically motivated diabetes prediction task. By partitioning data with a Dirichlet-based Non-IID scheme, we ensured that the experimental conditions reflect the demographic fragmentation inherent in real hospital networks rather than the artificial uniformity of IID benchmarks.

The federated pipeline matched centralized accuracy (70.91% vs. 69.48%) while guaranteeing that no patient record left its host site—a prerequisite for compliance with HIPAA and GDPR. More importantly, the results underscore that raw accuracy alone is an inadequate yardstick for medical machine learning: threshold tuning and localized feature engineering enabled both configurations to reach clinically relevant Recall levels. At the same time, our experiments offer a candid account of the cost of formal privacy: enabling DP-SGD at $\epsilon = 1.0$ reduced Recall from 73.70% to 52.22%, a penalty driven by the high noise-to-signal ratio on this small-parameter model. The privacy-utility curve in Table I gives practitioners a concrete reference for calibrating the privacy budget to their institution’s risk appetite. Looking ahead, the most impactful next step is to repeat these evaluations on larger, multi-modal clinical datasets. With richer feature spaces, the relative magnitude of DP noise diminishes, raising the prospect of strong formal privacy guarantees that no longer come at the expense of diagnostic safety.

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